SAFARICOM SEMA

Technical System Documentation & Engineering Manual

Intelligent Rapid Citizen Incident Ingestion & NOC Dispatch Operations

DOCUMENT SPECIFICATIONS & AUDIT PROFILE

System Name: Safaricom SEMA (Sauti ya Mwananchi)

Release Version: v2.1.0 Enterprise Production

Target Environment: Google Cloud Run & Safaricom Hybrid Network Stack

Document Classification: Technical Engineering Specification & SOP Manual

Authoring Unit: Safaricom Technology Architecture & NOC Escalations Desk

Executive Abstract

Safaricom SEMA is an enterprise-grade citizen incident reporting and automated dispatch engine designed for Safaricom PLC Kenya. The system eliminates communication friction by providing zero-barrier voice and text reporting across all 47 Kenyan counties in English, Kiswahili Sanifu, and Kenyan Sheng. Powered by Google Gemini 2.5 AI, real-time in-browser speech recognition, audio frequency analysis, and automated Network Operations Center (NOC) field engineer dispatch, SEMA cuts incident triage time from hours to seconds while tracking high-tension subscriber sentiment and enforcing strict SLA resolution milestones.

Document Organization

This manual serves system administrators, DevOps engineers, full-stack developers, NOC supervisors, and technical field officers. It is structured into 9 core operational sections covering architecture, setup, APIs, algorithms, SOPs, and troubleshooting.

Section Title	Reference	Primary Scope
1. System Architecture & Tech Stack	Section 1	Frontend, NLP, Backend & Cloud Infra
2. Setup, Installation & Environment Config	Section 2	Prerequisites, env vars, build pipelines
3. Customer Portal & Direct QR Engine	Section 3	Mobile deep-linking, touchless routing
4. Real-Time Voice Streaming Subsystem	Section 4	Web Speech API, Audio Visualizer, Gemini
5. Kenyan Multilingual NLP & Sentiment	Section 5	Sheng/Swahili lexicons, Frustration scoring
6. NOC Admin Dispatch & Email Automation	Section 6	Field officer dispatch notices, SLAs
7. Comprehensive REST API Reference	Section 7	All HTTP endpoints, schemas, payloads
8. Standard Operating Procedures (SOPs)	Section 8	SOP-01 to SOP-04 workflow guidelines
9. Troubleshooting, Security & Maintenance	Section 9	Microphone permissions, recovery, audit

1. System Architecture & Tech Stack

SEMA employs a modular, decoupled full-stack architecture built on modern web standards and Google Cloud services. The system is designed for high availability, sub-100ms UI responsiveness, and real-time offline-tolerant operation.

1.1 Layer-by-Layer Architectural Decomposition

- ï¿½ **Frontend Presentation Layer:** React 19, TypeScript, Tailwind CSS v4, Vite bundler, motion animation library, Lucide icon engine.
- ï¿½ **Speech & Audio Processing Layer:** HTML5 Web Speech API (SpeechRecognition for real-time continuous client-side (AudioContext + AnalyserNode for live soundwave visualizer), and SpeechSynthesis for accessibility feedback.
- ï¿½ **Intelligence & Multilingual NLP Layer:** Google Gemini 2.5 Flash multimodal model via @google/genai TypeScript SDK, morphological lexicons for instant zero-latency offline triage.
- ï¿½ **Backend Microservices Layer:** Node.js 20+ with Express REST API server, automated incident dispatch routing, SLA officer email dispatch generator.
- ï¿½ **Infrastructure & DevOps Layer:** Containerized Google Cloud Run execution, Docker containers, Cloud SQL persistence and GitHub Pages static synchronization.

1.2 Architectural Data Flow

```
[Citizen Device] -> (Camera QR Scan / Direct Web URL)
            %
            %ï¿½
[SEMA Customer Portal]
            % % %   1 .   M i c   I n p u t   - >   W e b   S p e e c h   A P I   ( C o n t i n u o u s   I n t e
            % % %   2 .   A u d i o   W a v e   - >   W e b   A u d i o   A n a l y s e r   ( R e a l - T i m e
            % % %   3 .   T e x t   E d i t   &   L a n g u a g e   H i n t   - >   C l i e n t - s i d e   n o r
            % % %   4 .   F o r m   S u b m i t   - >   P O S T   / a p i / s u b m i s s i o n s
                          %
                          %ï¿½
[Node.js Express API & Gemini AI Engine]
            % % %   L a n g u a g e   I d e n t i f i c a t i o n   ( E n g l i s h   /   K i s w a h i l i   /
            % % %   I s s u e   C a t e g o r i z a t i o n   ( H o m e   F i b r e ,   M - P E S A ,   4 G / 5 G
            % % %   F r u s t r a t i o n   &   H i g h - T e n s i o n   S e n t i m e n t   S c o r i n g   ( 0
            % % %   S L A   W i n d o w   C a l c u l a t i o n   ( 3 0 m ,   2 h ,   4 h ,   2 4 h )
            % % %   A u t o m a t e d   R e s p o n s e   T e a m   M a t c h i n g
                          %
                          %ï¿½
[Safaricom NOC Dispatch Console]
            % % %   I n t e r a c t i v e   R e a l - T i m e   T r i a g e   &   S t a t u s   D o t   F i l t e
            % % %   A u t o - A s s i g n   A l l   U n a s s i g n e d   ( 1 - C l i c k   B a t c h   A I   D i
            % % %   A u t o m a t e d   O f f i c e r   E m a i l   N o t i c e s   ( E n g .   D e n n i s   K i
            % % %   C S V   O p e r a t i o n s   A u d i t   E x p o r t
```

2. Setup, Installation & Configuration

SEMA can be deployed as a local development instance, a full-stack production container on Google Cloud Run, or a static client on GitHub Pages.

2.1 System Prerequisites

ï¿½ **Node.js Runtime:** Node.js v20.x or higher LTS (with npm v10+ or Bun).

ï¿½ **Browser Compatibility:** Google Chrome, Microsoft Edge, Safari 15+, or Firefox with microphone access enabled.

ï¿½ **Network Connectivity:** HTTPS protocol required in production to allow browser navigator.mediaDevices microphone

2.2 Environment Configuration

Create a `.env` file in the root workspace directory containing:

```
# SEMA Environment Configuration
PORT=3000
NODE_ENV=production
GEMINI_API_KEY=AIzaSy... # Optional: Google Gemini API key for live multimodal processing
VITE_PUBLIC_URL=https://ais-pre-zhdir2qsygkujxuwikuyds-576766090545.europe-west3.run.app
```

2.3 Build & Execution Commands

Command	Target Execution	Description
npm run dev	Development	Spawns Express API + Vite middleware on port 3000
npm run build	Production	Compiles React TypeScript into /dist and mirrors to /docs
npm start	Production	Launches production Node.js server from server.ts via tsx
npm run lint	Verification	Runs TypeScript strict type-checking across codebase

3. Customer Voice & Multilingual NLP Engine

One of SEMAs foremost innovations is its zero-barrier voice capture system, engineered to transcribe Kenyan accents and idioms accurately.

3.1 Continuous Speech Recognition Subsystem

SEMA leverages the browser SpeechRecognition API with explicit initial getUserMedia() permission negotiation to avoid iframe sandbox permission blocks. Transcription features include:

- ï¿½ **Continuous Streaming:** Speech is captured in real-time chunks without requiring the user to pause.
- ï¿½ **Interim Text Sync:** Interim transcriptions appear directly inside the textarea element with auto-scroll.
- ï¿½ **Dual Language Listeners:** Switches between en-KE (Kenyan English) and sw-KE (Kenyan Kiswahili) recognizers automatically.
- ï¿½ **Web Audio Visualizer:** Reads live frequency decibels to render 32 interactive green equalizer bars while the user speaks.
- ï¿½ **Gemini AI Fallback:** If the browser lacks SpeechRecognition (e.g. legacy browsers), recorded audio chunks are sent via /api/transcribe where gemini-3.5-transcribe extracts the text.

3.2 Kenyan Sheng & Kiswahili Linguistic Rules

SEMA incorporates a specialized token analysis dictionary containing over 50 Kenyan colloquial terms:

Language/Dialect	Sample Key Tokens	Routing Application
Kenyan Sheng	manze, rada, noma, chapaa, keja, bazenga, tutaisort,	Auto-identifies youth/informal reports; generates
Kiswahili Sanifu	shida, tatizo, stima, huduma, hospitali, mwananchi,	Applies national language acknowledgement;
Kenyan English	outage, router, optical, fiber, downtime, payment,	Directly extracts technical domain nouns for

3.3 Frustration & High-Tension Detection Scoring

Every incident receives a Frustration Score from 0 to 100 based on linguistic sentiment, emergency keywords (e.g., hospital discharge, medical paybill stuck, commercial link down), and outage duration. Reports scoring 70+ are automatically flagged as High Tension, triggering visual danger badges in the NOC console and expedited 30-minute / 2-hour SLAs.

4. NOC Admin Dispatch & Email Automation

The Network Operations Center (NOC) Admin Console provides real-time triage, interactive milestone tracking, automated AI team assignment, and email dispatch notices for field engineers.

4.1 Safaricom Response Teams & Designated Leads

Response Team Name	Lead Officer	Official Email Address	Target SLA
Home Fibre & Enterprise NOC	Eng. Dennis Kiprono	dennis.kiprono@safaricom.co.ke	2 Hours
M-PESA Escalations & Reversals	Officer Grace Mwangi	grace.mwangi@safaricom.co.ke	30 Mins
4G/5G BTS & RAN Mobile Team	Supervisor Brian Otieno	brian.otieno@safaricom.co.ke	4 Hours
SIM Swap & Cyber-Fraud Unit	Insp. Hassan Ali	hassan.ali@safaricom.co.ke	1 Hour
Customer Care & Airtime Billing	Coord. Agnes Muthoni	agnes.muthoni@safaricom.co.ke	24 Hours
Lipa Na M-PESA & Merchant Desk	Lead John Kimani	john.kimani@safaricom.co.ke	6 Hours

4.2 Automated Incident Dispatch Notices

When an incident is assigned (either manually or via the 1-Click AI Auto-Assign All button), the system generates an official dispatch notice sent to the assigned field officers email. The notice includes incident reference, SLA deadline, full original text, translation, coordinates, and diagnostic directives.

4.3 Incident Lifecycle Milestones

ï¿½ **Received (Amber):** Newly logged by citizen; awaiting NOC triage or initial engineering review.

ï¿½ **Assigned (Blue):** Field officer and response team assigned; dispatch notice transmitted.

ï¿½ **In Progress (Purple):** Field crew mobilized on-site; optical fiber splicing or switch query active.

ï¿½ **Resolved (Emerald):** Remedial action verified; customer link restored; SLA cleared.

5. Comprehensive REST API Reference

All endpoints accept and return JSON payloads and run on the primary server application.

5.1 Public Reporting Endpoints

ï¿½ **POST /api/submissions:** Submit an incident report. Accepts { text, countyOrWard, contactInfo, captureMethod }. Returns referenceCode, detectedLanguage, priority, and aiCommitmentResponse.

ï¿½ **POST /api/transcribe:** Transcribe voice audio chunk. Accepts { audioBase64, mimeType, languageHint, clientTranscript }. detectedLanguage, englishTranslation }.

ï¿½ **POST /api/transform-text:** Refine or translate text. Accepts { text, targetMode: "polish"|"to_en"|"to_sw"|"to_sheng" }. Returns {

ï¿½ **POST /api/sample-voice:** Fetch sample audio with Kenyan accent. Accepts { language: "en"|"sw"|"sheng" }. Returns sample audio.

5.2 NOC Dispatch & Management Endpoints

ï¿½ **GET /api/submissions:** Fetch all logged incident reports with current status, priority, and assigned team.

ï¿½ **POST /api/submissions/:id/ai-classify:** Execute Gemini AI classification for a single incident to generate category, urgency,

ï¿½ **PATCH /api/submissions/:id/assign:** Assign an incident to a response team and officer. Body: { assignedTeam, sendEmailAlert }. Returns updated record and created email alert.

ï¿½ **POST /api/submissions/:id/send-email-alert:** Manually dispatch or re-send dispatch email alert notice to the assigned officer.

ï¿½ **POST /api/submissions/auto-assign-all:** Marquee batch operation: Triages all unassigned reports using Gemini AI, assigns notices.

ï¿½ **PATCH /api/submissions/:id/status:** Quick status update for an incident (e.g. Received -> In Progress -> Resolved). Body: {

ï¿½ **GET /api/email-alerts:** Fetch all dispatched officer email audit records.

6. Standard Operating Procedures (SOPs)

SOP-01: Citizen Voice Reporting Protocol

Purpose: Ensure high-fidelity capture of citizen network or payment issues without typing barriers.

- ï¿½ **Step 1:** Citizen scans SEMA QR Code on poster, router box, or bill. The device immediately opens https://<domain>/?view=customer#customer-portal.
- ï¿½ **Step 2:** Citizen clicks "Tap to Speak (Mic)" and speaks naturally in English, Kiswahili, or Sheng.
- ï¿½ **Step 3:** Real-time transcript streams into the text box. The user verifies the text, selects county/ward, and clicks "Submit Report".
- ï¿½ **Step 4:** System generates SEMA-SAF reference code and instant localized AI commitment response.

SOP-02: NOC Rapid Triage & AI Batch Auto-Assignment

Purpose: Prevent backlog and mobilize field teams within 5 minutes of incident logging.

- ï¿½ **Step 1:** NOC Controller accesses Dispatch Console at #admin.
- ï¿½ **Step 2:** Reviews unassigned queue metric counter. Clicks "AI Auto-Assign All" marquee button.
- ï¿½ **Step 3:** Gemini AI evaluates all pending reports, calculates urgency SLAs, maps to specialized teams, and dispatches email team leads.
- ï¿½ **Step 4:** NOC Controller verifies high-tension (red border) items and contacts field leads directly if required.

SOP-03: Field Engineering Remediation & SLA Clearance

Purpose: Verify on-site technical resolution and update central records.

- ï¿½ **Step 1:** Field engineer receives dispatch notice email on mobile with incident coordinates and customer problem statement.
- ï¿½ **Step 2:** Engineer opens dispatch link, sets status to "In Progress", and conducts physical repair (e.g. fiber OTDR re-splicing or reset).
- ï¿½ **Step 3:** Once optical power or network carriers are verified within specification, engineer sets status to "Resolved" and logs notes.

7. Troubleshooting, Security & Reliability

7.1 Troubleshooting Matrix

Observed Issue	Root Cause	Automated Mitigation / Resolution
Microphone Access Denied	Browser blocked mic permission or	Call explicit navigator.mediaDevices.getUserMedia
SpeechRecognition Not Found	Firefox or older browser without	App falls back to client textarea editing and Gemini
QR Code Resolving to Localhost	Phone camera scanning localhost URL	getScannableWebUrl() dynamically substitutes public
Gemini API Rate Limit / No Key	Missing or exhausted GEMINI_API_KEY	Autonomous fallback to built-in rule-based classifier and

7.2 Security & Data Privacy Protocols

ï¿½ **Anonymity by Default:** Citizen phone numbers and contacts are strictly optional; anonymous civic reporting is fully supported.
ï¿½ **No Secret Storage in Frontend:** All Gemini AI calls and email dispatch configurations reside on the backend server.
ï¿½ **Sanitized Git Configuration:** PATs and deployment credentials are removed immediately after deployment sync.
ï¿½ **Data Sanitization:** Submissions strip malicious HTML/script injections before storage and rendering.

7.3 Version History & Release Notes

ï¿½ **v1.0.0 (Launch):** Core citizen text reporting, basic admin table, and rule-based categorization.
ï¿½ **v1.2.0 (Intelligence):** Integrated Google Gemini 2.5 AI, Sheng linguistic model, and interactive SLA tracking.
ï¿½ **v1.5.0 (Voice & QR):** Added real-time Web Speech transcription, audio wave visualizer, and dynamic QR code.
ï¿½ **v2.1.0 (Current):** High-tension sentiment scoring, automated field engineer email routing, and one-click PDF technical manual